

Agenda

1. Announcements

- no HW due today
- NO discussion quiz tomorrow
- Schedule changes:
 - 9.5 HW due March 22
 - Module Quiz 20 due MONDAY March 20
- Exam 1 will be returned next Thursday

2. Section 9.5

Section 9.5 - Linear Differential Equations

not separable, but solvable

Example: $x^3 \frac{dy}{dx} + 3x^2 y = e^x$

↑ ↑ ↑ ↑
First derivative der. second
term term first term
Product Rule

implicit differentiation
 $\frac{d}{dx}(x^3 y) = x^3 \frac{dy}{dx} + 3x^2 y$

$$\frac{d}{dx}(x^3 y) = e^x$$

$$\int \frac{d}{dx}(x^3 y) dx = \int e^x dx$$

$$x^3 y = e^x + C$$

$$y = \frac{e^x + C}{x^3}$$

Activity: $e^{x^3} \frac{dy}{dx} + 3x^2 e^{x^3} y = 6x^2 e^{x^3}$

$\Rightarrow \int \frac{d}{dx}(e^{x^3} y) = \int 6x^2 e^{x^3}$
 $u = x^3$
 $du = 3x^2$

$$e^{x^3} y = 2e^u + C$$

$$e^{x^3} y = 2e^{x^3} + C$$

$$y = \frac{2e^{x^3} + C}{e^{x^3}}$$

Example: $y' + y = 3$

The LHS is not a product rule. *integrating factor*

It is if we multiply by e^x .

How did I know to multiply by e^x

$$e^x(y' + y) = e^x(3)$$

$$\frac{d}{dx}[e^x y] = 3e^x$$

$$e^x y = \int 3e^x$$

$$\Rightarrow \boxed{\frac{3e^x + C}{e^x} = y}$$

A linear equation ^{that} is a first-order differential equation that can be expressed as

$$\frac{dy}{dx} + P(x)y = Q(x) \quad | \quad \begin{matrix} \text{ex.)} \\ y' + 1y = 3 \\ P(x)=1 \quad Q(x)=3 \end{matrix}$$

Where $P(x)$ and $Q(x)$ are continuous functions on an interval.

$$\frac{dy}{dx} + P(x)y = Q(x)$$

Activity: For each equation, determine whether it is separable, linear, neither, or both.

1. $\frac{dy}{dx} + 3y = e^{2x}$

linear

2. $\frac{dy}{dt} = 2 - ty$ $\frac{dy}{dt} + ty = 2$

linear

3. $y' + \overset{P(x)}{x}y = \overset{Q(x)}{x}$

$y' = x(1-y)$

Both

4. $xy' = y + x^2 \sin x$

$y' = \frac{1}{x}y + x \sin x$

$y' - \frac{1}{x}y = x \sin x$ linear

5. $e^t \frac{dy}{dt} + y^2 = y$

$\frac{dy}{dt} = (y - y^2)e^{-t}$

Seperable

6. $x^2 \frac{dy}{dx} + e^x y^2 = 1 - 3xy$

Neither

Solving Linear Equations

For ALL linear equations:

Suppose $\frac{dy}{dx} + P(x)y = Q(x)$ is a linear differential equation,

We define $I(x) = e^{\int P(x)dx}$, where $\int P(x)dx$ is a particular antiderivative, as the Integrating factor.

$+C \Rightarrow +O$
do not write.

To solve:

① multiply both sides by $I(x)$

$$e^{\int p(x) dx} \frac{dy}{dx} + p(x) e^{\int p(x) dx} y = Q(x) e^{\int p(x) dx}$$

IBP commonly arises here

② combine LHS with product rule

$$\frac{d}{dx} \left[e^{\int p(x) dx} y \right] = Q(x) e^{\int p(x) dx}$$

③ Integrate:

④ solve for y

$$e^{\int p(x) dx} y = \int Q(x) e^{\int p(x) dx}$$

$$I(x) = \int Q(x) I(x) dx$$

Example: $\frac{dy}{dx} + y = \cos(e^x)$

$$\int p(x) dx = x$$
$$I(x) = e^x$$

$$\textcircled{1} \quad e^x \frac{dy}{dx} + e^x y = e^x \cos(e^x)$$

$$\textcircled{2} \quad \frac{d}{dx} \left[\underbrace{e^x}_{I(x)} y \right] = e^x \cos(e^x)$$

$$\textcircled{3} \quad e^x y = \int e^x \cos(e^x) dx$$

$$= \sin(e^x) + C$$

$$y = \frac{\sin(e^x) + C}{e^x}$$

$$u = e^x$$
$$du = e^x dx$$

$$\frac{du}{e^x} = dx$$

Activity: $t \frac{dy}{dt} + 4y = t^5$, for $t > 0$

Example: $\frac{dv}{dt} - 2tv = 3t^2e^{t^2}$ $v(0) = 6$